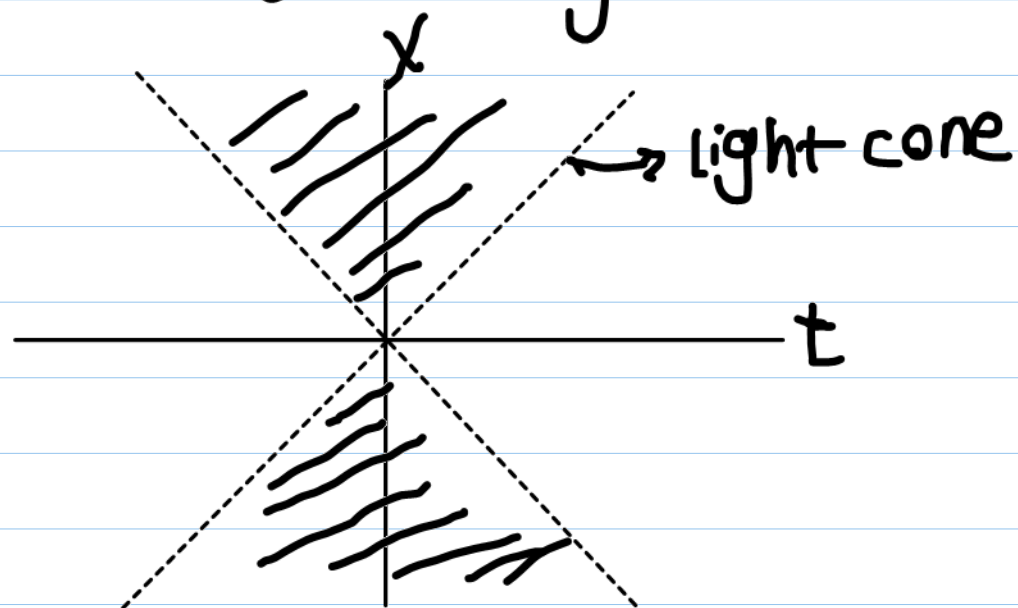


$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$$



$$\begin{aligned} \eta_{\mu\nu} \alpha^\mu \alpha^\nu &= (\alpha^0)^2 - (\alpha^1)^2 \quad (1D \text{ space}) \\ &= \left(\frac{dx^0}{d\tau} \right)^2 - \left(\frac{dx^1}{d\tau} \right)^2 = \gamma^2 \left(\left(\frac{dx^0}{dt} \right)^2 - \left(\frac{dx^1}{dt} \right)^2 \right) = -a^2 \end{aligned}$$

a : acceleration

u : velocity

t : the time that is measured by the non-inertial observer

τ : the time that is measured by the inertial observer

$$u^\mu = \frac{dx^\mu}{d\tau} \quad U^\mu = (c, \vec{u})$$

$$\frac{dt}{d\tau} = \gamma = \frac{1}{\sqrt{1 - \frac{u^2}{c^2}}}$$

$$\begin{aligned} \textcircled{*} \text{ In general, } \alpha^\mu &= \frac{dx^\mu}{d\tau} = \gamma \frac{dx^\mu}{dt} = \gamma \left(c \frac{d\gamma}{dt}, \frac{d(\gamma \vec{u})}{dt} \right) \\ &= (\alpha^0, \vec{\alpha}) \end{aligned}$$

then, in 1D $\gamma^2 = 1 - \left(\frac{d\mathbf{r}}{dt}\right)^2$

But how to prove it?

For $\frac{d\mathbf{r}}{dt}$ and $\beta = \frac{v}{c} \Rightarrow \gamma = (1 - \beta^2)^{-1/2}$

(i) $\frac{d\gamma}{dt} = \frac{-1}{2} (1 - \beta^2)^{-3/2} \cdot (-2\beta) \frac{d\beta}{dt} \Rightarrow \frac{d\gamma}{dt} = (1 - \beta^2)^{-3/2} \beta \frac{d\beta}{dt}$

$\Rightarrow (1 - \beta^2)^{-3/2} = \gamma^3 \Rightarrow \frac{d\gamma}{dt} = \gamma^3 \beta \frac{d\beta}{dt}$

$\beta = \frac{v}{c} \Rightarrow \frac{d\beta}{dt} = \frac{1}{c} \frac{dv}{dt} = \frac{a}{c} \Rightarrow \frac{d\gamma}{dt} = \gamma^3 \frac{v}{c} \frac{a}{c}$

$\frac{d\gamma}{dt} = \gamma^3 \frac{va}{c^2} //$

(ii) $\frac{d(\gamma v)}{dt} = \gamma \frac{dv}{dt} + v \frac{d\gamma}{dt} = \gamma a + v \left(\gamma^3 \frac{va}{c^2} \right) =$

$\gamma a \left(1 + \gamma^2 \frac{v^2}{c^2} \right) //$

$1 - \gamma^2 = 1 - \frac{1}{1 - \frac{v^2}{c^2}} = 1 - \frac{1}{\frac{c^2 - v^2}{c^2}}$

$\Rightarrow 1 - \frac{c^2}{c^2 - v^2} = 1 + \frac{c^2}{v^2 - c^2} \Rightarrow \frac{v^2 - c^2 + c^2}{v^2 - c^2} = \frac{v^2}{v^2 - c^2} = \frac{v^2 (1)}{c^2 (v^2 - c^2)}$

$= - \frac{v^2}{c^2} = 1 - \gamma^2 \Rightarrow \gamma^2 = 1 + \gamma^2 \frac{v^2}{c^2} //$

$\Rightarrow \gamma a \left(1 + \gamma^2 \frac{v^2}{c^2} \right) = \gamma a (\gamma^2) = \gamma^3 a //$

$(i)^2 - (ii)^2 = ?$

$$\partial^\mu [c^\mu (\dot{v})^2 - (\dot{x})^2] = \frac{d\gamma}{dt} - \frac{d(\gamma v)^2}{dt} = 0$$

$$\partial^\mu [c^\mu (\gamma^3 \frac{v}{c^2}) - (\gamma^3 \dot{x})] = [\gamma^6 \frac{v}{c^2} \partial^\mu - \gamma^6 \partial^\mu] \gamma^3$$

$$\gamma^3 \gamma^6 \partial^\mu (\frac{v}{c^2} - 1) = -\gamma^6 \partial^\mu \gamma^{-2} = -\partial^\mu \gamma^6 =$$

$$(\alpha^0)^2 - (\alpha^1)^2 = -\partial^\mu \gamma^6, (\alpha^1)^2 - (\alpha^0)^2 = +\alpha^2$$

$$\Rightarrow -\partial^\mu \gamma^6 = +\alpha^2 \Rightarrow d\gamma^3 = +\alpha //$$

$$\alpha = \frac{d(\gamma v)}{dt} \Rightarrow \alpha dt = d\left(\frac{v}{\sqrt{1-\frac{v^2}{c^2}}}\right)$$

$\alpha > 0$ assume

$$v(t_0) = 0$$

$$\Rightarrow v(t) = \frac{\alpha t}{\sqrt{1 + (\frac{\alpha t}{c})^2}}$$

What is that $x(t)$?

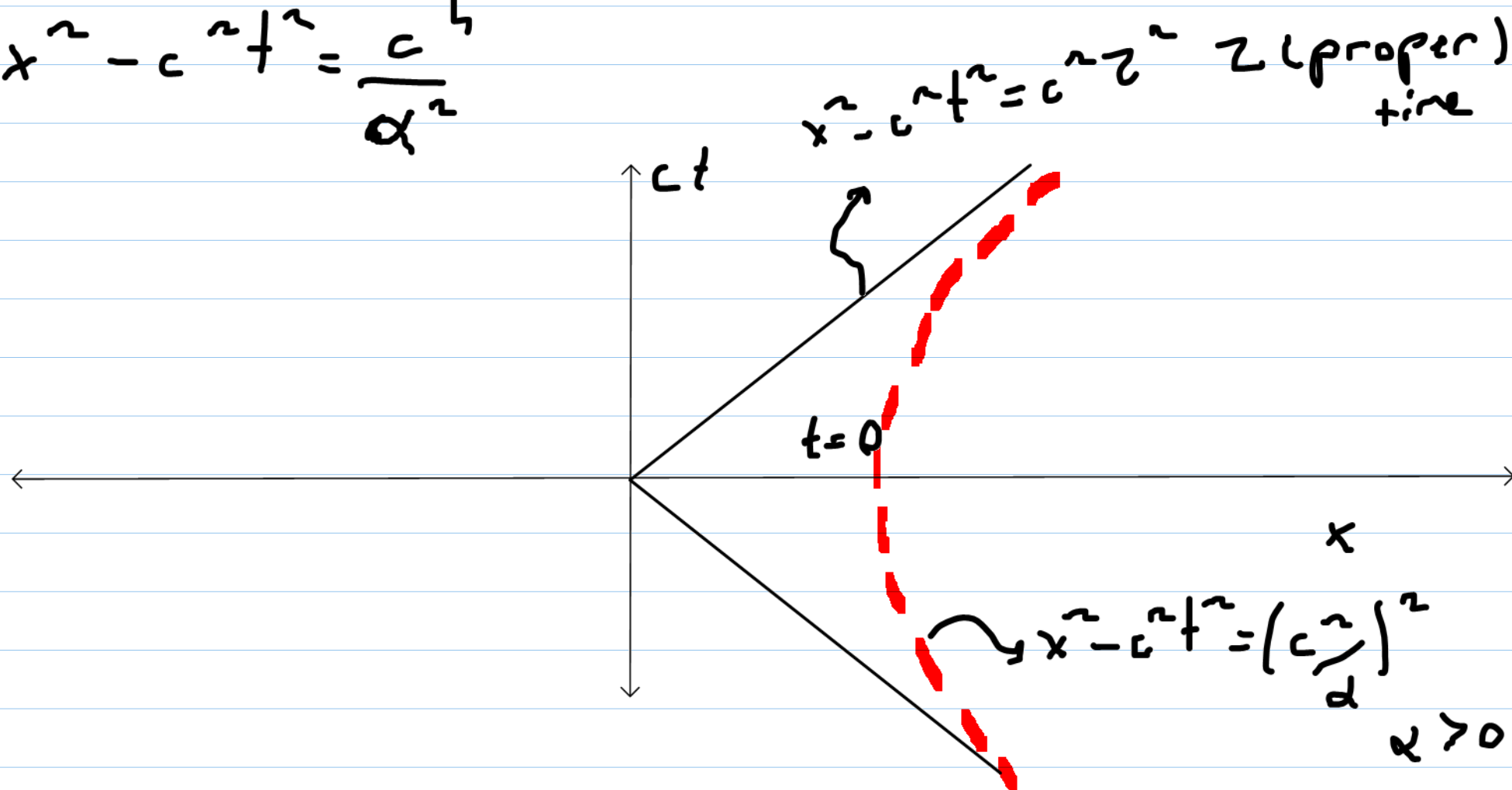
$$x(t) = \int_0^t \frac{\alpha t'}{\sqrt{1 + \left(\frac{\alpha t'}{c}\right)^2}} dt'$$

$$(35) \int \frac{x}{\sqrt{x^2 \pm a^2}} = \sqrt{x^2 \pm a^2}$$

$$\Rightarrow \frac{c^2}{\alpha} \sqrt{1 + \left(\frac{\alpha t}{c}\right)^2}$$

$$\Rightarrow x^2 = \frac{c^4}{\alpha^2} \left(1 + \frac{\alpha^2 t^2}{c^2}\right) = \frac{c^4}{\alpha^2} + c^2 t^2$$

$$\Rightarrow x^2 - c^2 t^2 = \frac{c^4}{\alpha^2}$$



Note $\underbrace{\frac{x^2}{a^2} - \frac{y^2}{b^2}}_{\text{hyperbole}} = 1 \Rightarrow y = \pm \frac{b}{a} x \text{ asympt.}$

$$\underbrace{\frac{x^2}{c^4/\alpha^2}}_{\text{Inf limite}} - \frac{t^2}{c^2/\alpha^2} = 1 \Rightarrow x = \pm ct \text{ asympt}$$

arrives light cone

